PROMPT ENGINEERING

EX-05:Comparative Analysis of Different types of Prompting Patterns with Various Test Scenarios

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Aim

To evaluate and compare how different prompting strategies – naive prompts (vague or unstructured) versus basic prompts (precise and well-defined) – influence the outputs generated by an AI language model across multiple test scenarios. Analyze the differences in quality, accuracy, and depth of the generated responses.

Definition of Prompt Types

Naive Prompt

A naive prompt is a vague, broad, or unstructured question that provides little context or guidance to the AI model. It does not specify the format, depth, or focus of the expected answer. As a result, the model must make many assumptions, which often leads to generic or unfocused responses.

Example:
Tell me about space exploration.

Basic Prompt

A basic prompt is a clear, detailed, and structured prompt that provides specific instructions, context, and format guidance to the AI model. It minimizes ambiguity and helps the model produce a more focused, accurate, and practically useful response.

Example:
Summarize the key milestones of space exploration from 1957 to 2000. Use a timeline format with one-line descriptions. Limit the response to 150 words, suitable for a high school student.

Test Scenarios

Five test scenarios were selected to compare naive and basic prompts. For each scenario, both prompt types were submitted to the AI model and the responses were evaluated based on Quality, Accuracy, and Depth.

- Scenario 1 – Generating a Creative Story
- Scenario 2 – Answering a Factual Question
- Scenario 3 – Summarizing a Concept
- Scenario 4 – Providing Advice or Recommendation
- Scenario 5 – Explaining a Technical Topic

Scenario 1 – Generating a Creative Story

Naive Prompt

Prompt:
Write a story.

ChatGPT Response Summary:

ChatGPT generated a short and generic story about a boy finding a treasure chest on a beach. The characters had no depth, the plot was predictable, and the setting was vague. The story lacked a specific theme, emotional appeal, or originality.

Basic Prompt

Prompt:
Write a short creative story (around 150 words) about a 14-year-old boy who discovers an old radio in his grandfather's attic that can receive messages from the past. The story should have a surprising emotional ending and a tone that is nostalgic and mysterious. Write it in third person.

ChatGPT Response Summary:

ChatGPT produced a well-structured story with a distinct setting in a dusty attic, a named protagonist named Arjun, and a rising sense of mystery as the radio crackled to life. The surprise ending revealed that the last message was from his grandfather recorded on the day Arjun was born. The story maintained a consistent nostalgic tone and stayed within the word limit.

Evaluation:

Criterion	Naive Prompt	Basic Prompt

Quality	Low – generic and flat	High – emotional and well-crafted
Accuracy	Not applicable for creative tasks	Followed all given instructions
Depth	Shallow – no character or plot development	Good depth with a clear arc and twist

Scenario 2 – Answering a Factual Question

Naive Prompt

Prompt:
What is the water cycle?

ChatGPT Response Summary:

ChatGPT gave a one-paragraph textbook-style explanation stating that the water cycle involves evaporation, condensation, and precipitation. The answer was technically correct but brief, with no diagram description, no examples, and no explanation of why the water cycle matters to daily life.

Basic Prompt

Prompt:

Explain the water cycle to a Class 7 student. Include the four main stages (evaporation, condensation, precipitation, and collection), the role of the sun as the energy source, and a simple real-life example of how rain forms. Keep the answer under 200 words.

ChatGPT Response Summary:

ChatGPT gave a structured explanation that clearly defined each stage of the water cycle in simple language, mentioned that solar energy drives evaporation, and connected it to a relatable example of how puddles on a road disappear on a sunny day and eventually return as rain. The response was well-paced and age-appropriate.

Evaluation:

Criterion	Naive Prompt	Basic Prompt
Quality	Moderate – correct but incomplete	High – complete and well-structured
Accuracy	Accurate but missing key details	Accurate with stages and real example
Depth	Surface-level only	Good depth suited to the target audience

Scenario 3 – Summarizing a Concept

Naive Prompt

Prompt:

Summarize blockchain

ChatGPT Response Summary:

ChatGPT gave a lengthy and general explanation of blockchain covering distributed ledgers, cryptography, consensus mechanisms, and various applications across multiple paragraphs. The response read more like a technical article than a summary and was not easy to use as a quick reference.

Basic Prompt

Prompt:
Write a 5-sentence summary of blockchain technology for someone with no technical background. Avoid jargon. Focus only on what it is, how it works in simple terms, and one real-world application.

ChatGPT Response Summary:

ChatGPT produced a clear 5-sentence summary. It defined blockchain as a shared digital record book that no single person controls, compared it to a class register that every student can see but no one can secretly erase, and used the example of cryptocurrency payments as a real-world application.

Evaluation:

Criterion	Naive Prompt	Basic Prompt
Quality	Low for a summary – too long	High – concise and targeted
Accuracy	Accurate but unfocused	Accurate and well-focused
Depth	Too broad for a summary	Appropriate depth for a general audience

Scenario 4 – Providing Advice or Recommendation

Naive Prompt

Prompt:
How do I study better?

ChatGPT Response Summary:

ChatGPT provided a generic list of study tips such as taking notes, avoiding distractions, revising regularly, and sleeping well. The advice was broadly correct but not personalized. There was no structure indicating which tips were most important or how to apply them in practice.

Basic Prompt

Prompt:
I am a second-year B.Tech student who finds it hard to focus during self-study hours at night due to phone distractions, and I struggle to retain concepts from subjects like Data Structures and Algorithms. Give me 5 practical and specific study strategies that can help me improve focus and retention. Format the response as a numbered list with a one-line explanation for each strategy.

ChatGPT Response Summary:

ChatGPT provided 5 targeted strategies: using the Feynman Technique to explain DSA concepts aloud in simple words, keeping the phone in another room during a 45-minute focused study block, solving one LeetCode problem per day to reinforce algorithmic thinking, using spaced repetition with flashcards for syntax and definitions, and reviewing the previous day's notes for 10 minutes before starting a new topic. Each tip came with a practical explanation.

Evaluation:

Criterion	Naive Prompt	Basic Prompt
Quality	Low – too vague and generic	High – specific and immediately actionable
Accuracy	Correct but not useful without context	Highly relevant to the student's exact problem
Depth	Shallow – no personalization	Deep – addressed the exact subject and habit issue

Scenario 5 – Explaining a Technical Topic

Naive Prompt

Prompt:

What is cloud computing*?*

ChatGPT Response Summary:

ChatGPT gave a standard encyclopedia-style answer covering the definition of cloud computing, deployment models (public, private, hybrid), service types (IaaS, PaaS, SaaS), and several enterprise applications. The response was several paragraphs long and assumed a moderate level of technical knowledge.

Basic Prompt

Prompt:

Explain what cloud computing is to a 12-year-old child using a fun and simple analogy. Then give two examples of cloud computing that the child already uses in everyday life without realizing it. Keep the total response under 100 words.

ChatGPT Response Summary:

ChatGPT explained cloud computing using the analogy of renting a locker at school instead of carrying everything in your own bag – the data is stored somewhere safe and accessible whenever needed. It then gave two relatable examples: saving photos in Google Photos and playing a game on a browser without downloading it. The response was under 100 words and had a friendly, engaging tone.

Evaluation:

Criterion	Naive Prompt	Basic Prompt
Quality	Moderate – correct but too technical	High – perfectly matched to the audience
Accuracy	Accurate but not audience-appropriate	Accurate and appropriately simplified
Depth	Too deep for a general audience	Right depth for a child audience

Overall Comparison Table

Each response is rated on a scale of 1 to 5 for Quality, Accuracy, and Depth. (1 = Very Poor, 2 = Poor, 3 = Average, 4 = Good, 5 = Excellent)

Scenario	Quality (Naive)	Quality (Basic)	Accuracy (Naive)	Accuracy (Basic)	Depth (Naive)	Depth (Basic)
Creative Story	2	5	3	5	2	4
Factual Question	3	5	3	5	2	4
Summarizing	2	5	3	4	2	4

Advice	2	5	2	5	2	5
Technical Topic	3	5	3	5	3	5

Analysis

Does the AI consistently perform better with basic prompts?

Yes. Across all five test scenarios, the AI produced noticeably better responses when given basic prompts. The outputs were more focused, appropriately structured, and better matched to the intended audience and purpose. Naive prompts consistently produced responses that were either too generic, too long, or poorly tailored to the user's actual need.

Are there scenarios where naive prompts work equally well?

For simple factual questions where only a basic definition is required, naive prompts can still generate a correct and usable answer. For instance, asking "What is the water cycle?" does produce an accurate response. However, even in these cases, the basic prompt delivered a more structured, audience-appropriate, and practically useful answer.

Key Findings

- Clarity in prompts directly improves the quality, accuracy, and depth of AI responses across all types of tasks.
- Specifying the target audience (e.g., Class 7 student, 12-year-old child) significantly changes the tone and complexity of the response.
- Providing format instructions (e.g., numbered list, bullet points, word limit) makes the output immediately usable without further editing.
- Naive prompts force the AI to make assumptions, which often leads to responses that miss the user's actual intent.
- Basic prompts reduce the need for follow-up questions or re-prompting, saving time and improving overall productivity.
- Providing personal context (e.g., subject, specific struggle) leads to responses that are tailored and actionable rather than generic.

How to Structure Prompts for Optimal Results

Element	What to Include	Example
Role / Audience	Who is the response for?	Explain to a Class 7 student
Task	What exactly should the AI do?	Write a 5-sentence summary
Context	Background information or constraints	I am a second-year B.Tech student
Format	How should the output look?	Use a numbered list with one-line explanations
Length	How long should the response be?	Keep the answer under 150 words

Conclusion

This experiment clearly demonstrates that the quality of a prompt has a direct and significant impact on the usefulness of an AI-generated response. Basic prompts that specify the audience, format, length, and context consistently outperform naive prompts across all five test scenarios. Crafting well-structured prompts is an essential skill for anyone using AI tools effectively in academic, professional, or creative contexts.